

If I turned on the input from both eyes, I asked Dr. Ruggiero, wouldn't I see double? Isn't this why I suppressed in the first place? This concern is one of the reasons why it is so important to undertake vision therapy with an optometrist highly trained in binocular vision. To prevent this situation, Dr. Ruggiero gave me several procedures that taught me how to aim both eyes precisely at the same place in space at the same time. As I mastered this skill, something remarkable happened. I learned to fuse images from the two eyes and achieved what I thought was impossible: I began to see in stereoscopic depth.

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THE SPACE BETWEEN

It must be repeated here that, before stereopsis is actually experienced by the patient, there is nothing one can do or say which will adequately explain to him the actual sensation experienced. . . . Once the patient has experienced this new sensation, he is only too anxious to use it again and again until it is surely and definitely established.

—*Frederick W. Brock*

For most people, seeing in stereoscopic depth happens effortlessly. When we look at an object, our brain automatically compares the images seen by the right and left eyes. If the position of the image on the right fovea is a little different from that of the image on the left fovea, the brain commands the eyes to turn in or out to minimize this disparity. When I looked at something before I acquired stereovision, the object cast an image on the fovea of one retina but on a noncorresponding, nonfoveal point of the other retina. The disparity between these two images was too great for me to automatically make corrective convergence or divergence eye movements. I needed to learn very consciously how to aim my two eyes. To do this, I needed feedback that showed me where my eyes were pointing in space so that I could redirect them. Fortunately, Dr. Ruggiero had a solution, a simple device invented by optometrist Frederick W. Brock.

Dr. Brock was born in Switzerland in 1899 and came to the United States in 1921 to attend the Columbia School of Optometry. He was an inventive and playful man. When, one summer, he found himself without an outdoor grill, he simply fashioned one out of a garbage can lid. When his daughter went to summer camp, he sent her letters peppered with little drawings and visual puns. His lifetime passion was helping people with their vision, and in the mid-1900s, he helped

many strabismics learn to see in stereo. Not content to test merely what his patients could not see, he performed many thoughtful experiments to determine what and how his patients did see. From these investigations, and with his typical resourcefulness, he built some of his own vision aids out of everyday items. In so doing, he taught us that you can learn a lot from a simple piece of string.

When I learned to use the “Brock string,” I received the feedback that I needed to know where my eyes were pointing and then to redirect them so that they were aiming simultaneously at the same point in space. During one vision therapy session, Laurie took a five-foot-long piece of ordinary string on which were threaded three brightly colored beads. She attached one end of the string to a doorknob and gave the other end to me. She told me to walk backward until the string was taut and then to put the free end of the string up to the bridge of my nose ([Figure 6.1](#)). She slid one of the beads along the string so it was only a few inches from my face. Then she asked me to look at the bead and tell her how many strings I saw.

I thought that Laurie had asked a pretty ridiculous question. I looked down the string at the bead, saw one string, and told her so. Laurie nodded and then, while gently moving the string up and down, asked me to look again at the bead and tell her what I saw. To my surprise, I momentarily saw two strings emerging in front of the bead and two strings receding from the bead as in [Figure 6.2](#).